One Phase-Exceptional Set for Every Atomic Prefix: A Dyadic Maximal Parseval Theorem on Determinant-Two Fibers

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Abstract

TPC-159 controls cumulative periodic-core sums outside a dyadic shadow of scales, while TPC-160 shows that a prefix cutoff places one variation atom exactly at its selected endpoint. We open a different route by averaging an additive phase. An elementary Rademacher–Menshov dyadic decomposition and Parseval prove an L^2 bound for the maximum over all fiber prefixes. Consequently one phase-exceptional set controls every atomic endpoint, including endpoints inside the scale shadow. The theorem is pointwise in the endpoint but averaged in phase; it does not control a specified phase such as zero. This is an actual-core phase-metric child, with a maximal L^2 analytic norm, of the bad-endpoint frontier; it is not the program's positive-$L2$ gate or the pointwise fixed-phase closure.

Keywords. maximal inequality; Rademacher–Menshov decomposition; additive phase; atomic prefix; determinant-two fiber.

Claim boundary. All prefixes are controlled simultaneously only outside an exceptional set of additive phases. The result does not imply control at $\alpha = 0$ or at a named physical phase. Avoiding the TPC-159 scale shadow by phase averaging is not a physical endpoint registry. No full H3 return, 1/400, prime-pair lower bound, or twin-prime theorem is proved.

1 Ordered fiber prefixes

Retain the determinant-two data

$$q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az),$$

and let ρ be any bounded function; in the inherited actual core it may be periodic. Order the positive fiber points

$$0 < t(z_1) < \cdots < t(z_L) \le T.$$

The z_j are consecutive integers and

$$L \le T/q + 1. \tag{1}$$

Put

$$b_j = c_{z_j}\rho(z_j), \qquad S_k(\alpha) = \sum_{j=1}^{k} b_j \mathrm{e}(-\alpha z_j),$$

where $\mathrm{e}(x) = \exp(2\pi i x)$, and define

$$D_L = 1 + \lceil \log_2 L \rceil, \qquad G_T(\alpha) = \frac{q}{T} \max_{1 \le k \le L} |S_k(\alpha)|.$$

2 The dyadic maximal lemma

Theorem 2.1 (Finite dyadic maximal Parseval)**.** *For arbitrary $b_1, \dots, b_L \in \mathbb{C}$,*

$$\boxed{\int_{\mathbb{T}} \max_{1\le k\le L} \left|\sum_{j=1}^{k} b_j \mathrm{e}(j\alpha)\right|^2 d\alpha \le D_L^2 \sum_{j=1}^{L} |b_j|^2.} \tag{2}$$

Proof. Pad the coefficient sequence by zeros to length $H = 2^{\lceil \log_2 L\rceil}$. At each of the D_L dyadic levels, partition $\{1, \dots, H\}$ into aligned blocks of the corresponding power-of-two length. The binary expansion of k decomposes every prefix $\{1, \dots, k\}$ into disjoint aligned blocks, with at most one block from each level.

Let $B_I(\alpha) = \sum_{j\in I} b_j \mathrm{e}(j\alpha)$. Cauchy–Schwarz on the at most D_L selected blocks gives, for every k,

$$\left|\sum_{j\le k} b_j \mathrm{e}(j\alpha)\right|^2 \le D_L \sum_{\text{all dyadic } I} |B_I(\alpha)|^2.$$

The right side is independent of k. At a fixed level the blocks partition the coefficient indices, and Parseval gives

$$\sum_{I \text{ at that level}} \int_{\mathbb{T}} |B_I(\alpha)|^2 d\alpha = \sum_{j=1}^{L} |b_j|^2.$$

There are D_L levels, proving (2). □

This is the finite dyadic maximal argument commonly associated with Rademacher–Menshov estimates; no convergence theorem is imported [3].

3 One phase set for all actual-core endpoints

Theorem 3.1 (Maximal direct-twist core theorem)**.** *For the determinant-two fiber above,*

$$\boxed{\int_{\mathbb{T}} G_T(\alpha)^2 d\alpha \le D_L^2 \|\rho\|_\infty^2 \left(\frac{q}{T} + \frac{q^2}{T^2}\right).} \tag{3}$$

Hence for every $\lambda > 0$,

$$\operatorname{meas}\{\alpha : G_T(\alpha) > \lambda\} \le \frac{D_L^2 \|\rho\|_\infty^2}{\lambda^2} \left(\frac{q}{T} + \frac{q^2}{T^2}\right). \tag{4}$$

Proof. The common exponent shift from z_j to j has unit modulus. Apply theorem 2.1, use

$$\sum_j |b_j|^2 \le \|\rho\|_\infty^2 L,$$

then use (1). Chebyshev gives (4). □

Corollary 3.2 (Every endpoint in a terminal shell)**.** *Outside the phase set in* (4)*, every fiber endpoint $U = t(z_k) \in [\theta T, T]$, $0 < \theta \le 1$, satisfies*

$$\frac{q}{U} |S_k(\alpha)| \le \theta^{-1}\lambda. \tag{5}$$

Proof. The numerator is one of the prefix sums in G_T, and $T/U \le \theta^{-1}$. □

The exceptional set in (4) is common to all k. Therefore corollary 3.2 includes endpoints lying in the dyadic scale shadow of TPC-159 [1]. It does not assert that those endpoints are good for a fixed phase.

4 Power envelope and exact status

If

$$q \le (\log X)^{\eta_0}, \qquad \sqrt{X} \le T \le X, \qquad \|\rho\|_\infty \ll 1,$$

then $D_L = O(\log X)$ and

$$\|G_T\|_{L^2(\mathbb{T})} \ll X^{-1/4}(\log X)^{1+\eta_0/2}. \tag{6}$$

This is a positive fixed-X power in a maximal phase-L^2 analytic norm. The exact program progress label is

PROVED_L1_ACTUAL_CORE_PHASE_METRIC_MAXIMAL_PREFIX.

axis	value
analytic_norm	L2_PHASE_MAXIMAL
program_positive_L2	false
fixed_atom	false

It is stronger than one-endpoint phase Parseval because the maximum is inside the integral, at the cost of D_L^2.

5 Relation to the atomic endpoint barrier

TPC-160 proves that a literal prefix cutoff has one Abel derivative atom and that continuous scale-exceptional density cannot decide whether the selected endpoint is bad [2]. The present theorem changes the averaging variable:

route	pointwise variable	exceptional variable
TPC-159/160	phase / multiplier	scale
TPC-169	all prefix endpoints	phase.

Thus TPC-169 creates a feasible phase-metric child of O161.bad_endpoint_pointwise_core. It does not close the original node, because nothing proves that a source-locked production phase avoids the exceptional phase set. In particular, (4) cannot be evaluated at $\alpha = 0$.

6 Reproducible audit

The standard-library certificate source-locks TPC-160 and TPC-167. It verifies the aligned dyadic decomposition, including at most one block of each size, for every prefix of a non-power-of-two fixture. It evaluates the maximal function on a finite Fourier grid as an implementation check; the analytic proof certifies the continuous integral. Executable bad-snapshot mutations reject deletion of the squared depth cost, confusion of terminal and small-endpoint normalization, and promotion from endpoint-pointwise to phase-pointwise. Hashes certify integrity only.

7 Conclusion

One common phase-exceptional set now controls every atomic prefix on the actual determinant-two core. This crosses the scale-shadow barrier in a genuinely different norm, with an explicit positive fixed-X phase-L^2 power. The remaining selector gap is honest and sharp. TPC-170 asks whether a fixed phase outside one null set can be followed along an entire prescribed sequence of cores and then returned through Abel summation.

References

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[3] Antoni Zygmund. *Trigonometric Series.* Cambridge University Press, third edition, 2002.